

Categories, Proofs, and Processes: Assignment 6

Wira Azmoon Ahmad

4 Mar 2024

Question 1

Claim 1.1. *h is monotone.*

Proof. Suppose $x, y \in P, x \leq y$. Then

$$\begin{aligned} h(x) &\leq \bigvee \{h(x), h(y)\} \quad (\text{by LUB}) \\ &= h\left(\bigvee \{x, y\}\right) \\ &= h(y) \quad (\text{since } y = \bigvee \{x, y\}) \end{aligned} \tag{1}$$

So h is monotone. □

Claim 1.2. *U has a left adjoint.*

Define the functor $F : \mathbf{Set} \rightarrow \mathbf{SL}$

$$F(X) = (\mathcal{P}(X), \subseteq)$$

$$F(f : X \rightarrow Y) : \mathcal{P}(X) \rightarrow \mathcal{P}(Y) := A \mapsto \{f(a) | a \in A\}$$

which is the covariant powerset functor for arrows, so functoriality is satisfied.

The functor preserves least upper bounds $\forall f : X \rightarrow Y$ in $\mathbf{Set}$,

$$\begin{aligned} F(f)\left(\bigvee S\right) &= F(f)\left(\bigcup_{A \in S} A\right) \\ &= \left\{f(a) \mid a \in \bigcup_{A \in S} A\right\} \\ &= \bigcup_{A \in S} \{f(a) | a \in A\} \\ &= \bigcup_{A \in S} F(f)(A) \\ &= \bigvee_{A \in S} F(f)(A) \\ &= \bigvee \{F(f)(A) | A \in S\} \end{aligned} \tag{2}$$

For each object $C \in \text{Ob}(\mathbf{Set})$, define the unit arrow $\eta_C : C \rightarrow UFC$

$$\eta_C(a) = \{a\}$$

From $f : A \rightarrow B$, we can construct an arrow in **SL**

$$\hat{f} : \mathcal{P}(A) \rightarrow B := A \mapsto \bigvee F(f)(A)$$

where $U(\hat{f}) = \hat{f}$. Then

$$\begin{aligned} (U(\hat{f}) \circ \eta_A)(a) &= \hat{f}(\{a\}) = f(a) \\ \forall a, f(a) &= U(\hat{f}) \circ \eta_A(a) \end{aligned}$$

So $f = U(\hat{f}) \circ \eta_A$.

Suppose there's another $g : \mathcal{P}(A) \rightarrow B$ such that $f = U(g) \circ \eta_A$. Then it also preserves least upper bounds where $\forall X \subseteq A$,

$$g(\bigvee \{X\}) = g(X) = \bigvee \{g(a) | a \in X\} = \bigvee F(f)(X)$$

so $g = \hat{f}$ is unique.

Question 2

Question 3

Question 4